

Problem Set 2

Due on Canvas at 12:00 noon Friday September 27

Instructions: **No partial credit will be awarded on quantitative questions.** All answers must be entered in the Problem Set 2 Quiz on Canvas. Note that it is **not possible to edit your responses** on Canvas after submitting. Best practice is to **complete the problem set on paper** and then transfer solutions to Canvas. You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a ten point scale across all problems in the problem set. Note that after submitting on canvas, you may see a score, but that does not include any points for any qualitative questions and is only a preliminary score. Grades will be finalized a few days after the due date.

Material covered: This problem set covers the production and perfect competition lectures of the course (Sept 12-Sept 17).

Problem 1. “Cobb-Douglas” Production of Soccer Balls

Soccer balls are produced in Bangladesh by using a mix of capital (K, machines) and labor (L, employees) according to the following production function:

$$Q = 2L^{3/4}K^{1/4}$$

where K and L are measured in hours (note: the “power rule” for derivatives applies even to fractional exponents¹). An hour of labor costs \$20 and an hour of capital costs \$40 (the currency is the Bangladeshi Taka, but we will use the \$ sign). Throughout you may verify your answers with Excel/Solver.

- a. What is the optimal level of labor to employ as a function of the amount of capital?
- b. Assume that the firm wishes to produce 230 balls per hour at minimum cost. Assuming it can freely vary its use of capital and labor, how much capital and labor should it employ? (You may round to the nearest whole number)
- c. What is the total cost of producing the 230 soccer balls each hour? (In Bangladeshi Taka, but again we use the symbol \$)
- d. What is the average cost of producing a soccer ball?
- e. Discuss your optimal production plan, making sense of the labor-to-capital ratio that you have solved for.

¹ If you’re unsure about your derivative calculation, feel free to verify at sites like <https://www.derivative-calculator.net>, where you would just need to rename “x” the variable you are differentiating with respect to.

- f. What is the marginal product of labor evaluated at the quantity of labor and capital you found in part b? (Use Solver if you want.)

- g. What is the marginal product of capital evaluated at the quantity of labor and capital you found in part b? (Use Solver if you want.)

- h. Now suppose that the level of capital is fixed at the level you identified in part a (i.e. you cannot adjust your capital usage in the short run), but you want to produce 300 units of output instead of 230. How much labor should you employ to do this? What has happened to average cost? What has happened to the marginal products of capital and labor relative to their costs? (Again use Solver as needed.) What is going on here?

Problem 2. E-Scooters. The e-scooter craze is sweeping the nation, with cities like Los Angeles, Chicago, and Atlanta overrun with dockless electric scooters that can be rented via an app. A New Haven startup, Scüt, is hoping to bring the craze to the Elm City. They have determined that their daily cost function (where quantity is in number of scooters placed around the city on a given day) is

$$TC(q) = 80 + 20q + 0.05q^2$$

Unfortunately for Scüt, the e-scooter business is perfectly competitive, and they may face rivals such as Lime, Bird, and even Uber and Lyft in the future.

(a) What is Scüt's supply function for scooters (for the range of prices where they are willing to supply any), **as a function of the market price p (which is daily revenue per scooter) for an e-scooter?**

Supply function: $Q_S(P) =$ _____

(b) If the current prevailing market price (daily revenue) in the short run is \$40 per day per scooter, how many scooters will Scüt supply, and what will their daily profit be?

Supplied Quantity: _____

Daily Profit: _____

(c) Scüt's success has attracted some attention, leading to a price war in New Haven. In the short-run, the fixed cost component of total costs is unavoidable. What is the price below which Scüt would stop placing scooters around the city in the short term?

Price _____

(d) In the long run, there is free entry, and indeed several competitors join the market with the same cost function as Scüt. What will be the long-run equilibrium price (daily revenue) per e-scooter?

Price _____